

解方程

$$x^3y - 4y\mathrm{d}x + x^4\mathrm{d}y = 0$$

配凑得到

$$4x^3y\mathrm{d}x + x^4\mathrm{d}y = 3x^3y + 4y\mathrm{d}x$$

令 $u = x^4y$

则

$$\begin{aligned} \mathrm{d}u &= \frac{3u}{x} + \frac{4u}{x^4}\mathrm{d}x \\ \Rightarrow \frac{\mathrm{d}u}{u} &= \left(\frac{3}{x} + \frac{4}{x^4} \right) \mathrm{d}x \\ u &= Cxe^{-\frac{4}{3x^3}} \end{aligned}$$